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Artificial Intelligence and Machine Learning

(6CS012)

Task 3 Report

Racist and Sexist Tweet Detection using RNN, BiLSTM, and GloVe Embeddings

Github: **https://github.com/ASMIT123445/AI-and-ML-final-accesment**

Name: Asmit Pradhan

Student Id: 2408991

Group: L6CG18

Supervisor: Mr Shiv Kumar Yadav

Module Leader: Simon Giri

Abstract

This report presents an end-to-end deep learning pipeline for binary text classification, specifically the detection of racist and sexist content in tweets. The dataset, assigned by Herald College Kathmandu for the 6CS012 module, is a heavily imbalanced binary classification problem where approximately 93% of tweets are labelled as non-hate and only 7% as hate speech. This severe imbalance made plain accuracy a misleading metric, so macro-averaged F1 score was adopted as the primary evaluation criterion throughout the study.

Three deep learning models were designed, trained, and evaluated. Model 1 is a Simple RNN with a trainable embedding layer. Model 2 is a Bidirectional LSTM with a trainable embedding layer. Model 3 is a Bidirectional LSTM using frozen pretrained GloVe embeddings trained on two billion tweets. All models incorporated class weighting during training to address the class imbalance, along with SpatialDropout1D, recurrent dropout, L2 regularisation, and EarlyStopping callbacks. A decision threshold tuning strategy was applied on the validation set to optimise F1 score for the minority hate class, and the best threshold was then used for final evaluation on a held-out test set.

The BiLSTM with GloVe embeddings achieved the strongest performance, with the highest macro F1 score, demonstrating the value of domain-relevant pretrained word vectors for Twitter-based classification tasks. The Simple RNN performed weakest due to its inability to capture long-range dependencies and its susceptibility to the vanishing gradient problem. A Gradio-based real-time prediction interface was also implemented, allowing users to input a tweet and receive an instant classification with confidence scores.

Table of Contents

1. Introduction

1.1 Background

The rapid growth of social media platforms has brought with it a significant increase in the spread of hate speech, including racist and sexist content. Automated detection of such content is an important application of natural language processing (NLP) and deep learning, with direct implications for platform moderation, public safety, and online wellbeing. Tweet classification is a particularly challenging NLP task due to the informal and abbreviated nature of Twitter language, the widespread use of slang, sarcasm, hashtags, and user mentions, and the high degree of contextual ambiguity present in short texts.

Recurrent Neural Networks and their variants, particularly Long Short-Term Memory networks, have become standard architectures for sequential text classification tasks. LSTMs address the vanishing gradient problem inherent in simple RNNs by introducing gating mechanisms that control the flow of information across time steps. Bidirectional LSTMs further improve performance by processing the input sequence in both forward and backward directions, capturing context from both sides of a word simultaneously.

1.2 Objectives

The objectives of this project are as follows:

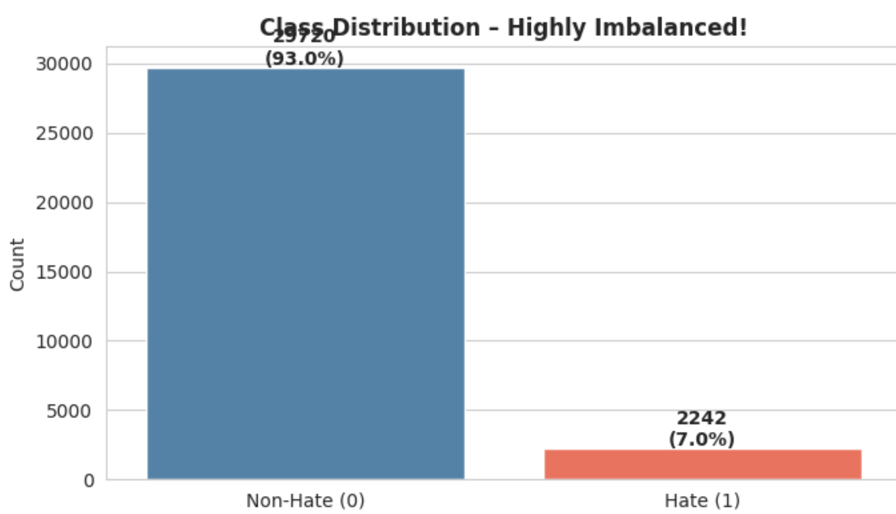
- To build a complete NLP preprocessing pipeline for raw tweet data including contraction expansion, text normalisation, stopword removal, and lemmatisation.
- To design and train three text classification models: a Simple RNN, a Bidirectional LSTM with trainable embeddings, and a Bidirectional LSTM with pretrained GloVe embeddings.
- To address class imbalance through class weighting and decision threshold tuning optimised for macro F1 score.
- To evaluate and compare all three models using accuracy, precision, recall, F1 score, and confusion matrices.

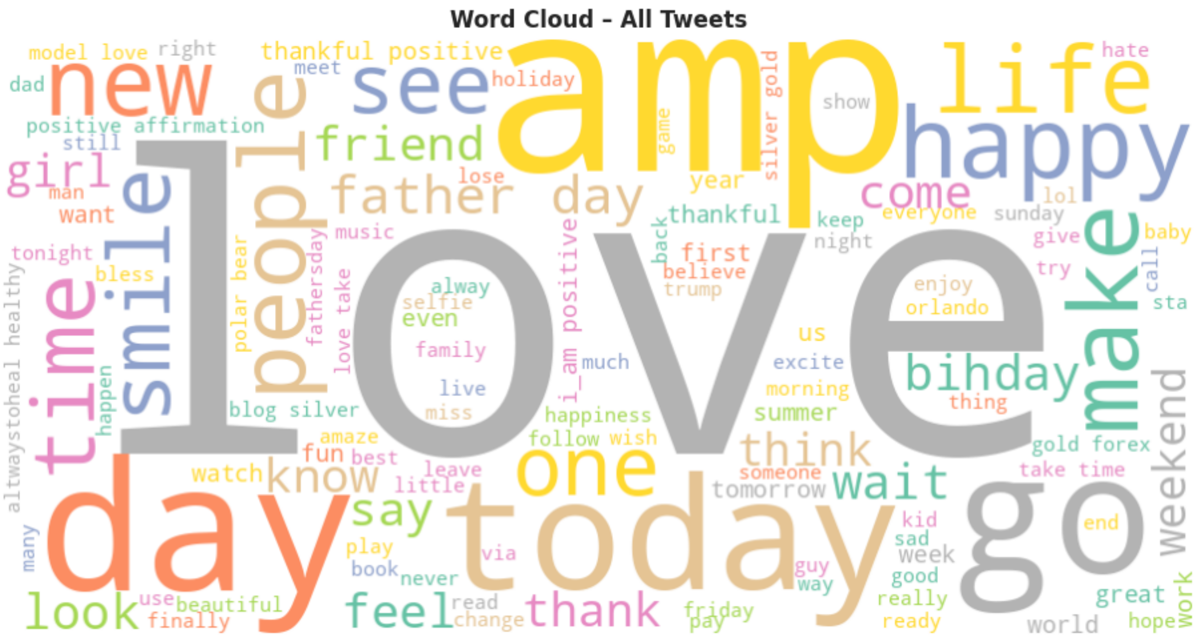
- To conduct error analysis on misclassified examples and identify the linguistic challenges behind prediction failures.
- To deploy a real-time prediction interface using Gradio.

2. Dataset

2.1 Description

The dataset used in this project is the Racist/Sexist Tweet dataset assigned by Herald College Kathmandu. It is a binary classification dataset where each tweet is labelled as either non-hate (label 0) or hate speech including racist and sexist content (label 1). The dataset was loaded from a CSV file containing at minimum a tweet column with raw text and a label column with binary class annotations. The labelled training file was split internally into training, validation, and test sets, as the separate test CSV provided with the dataset contained no labels and was therefore unusable for metric evaluation.

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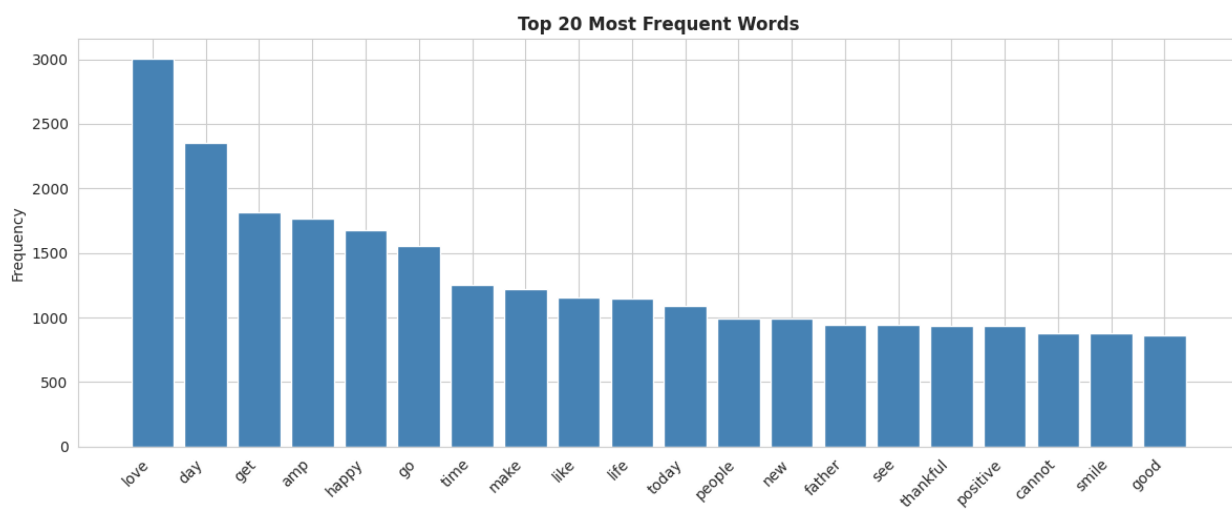
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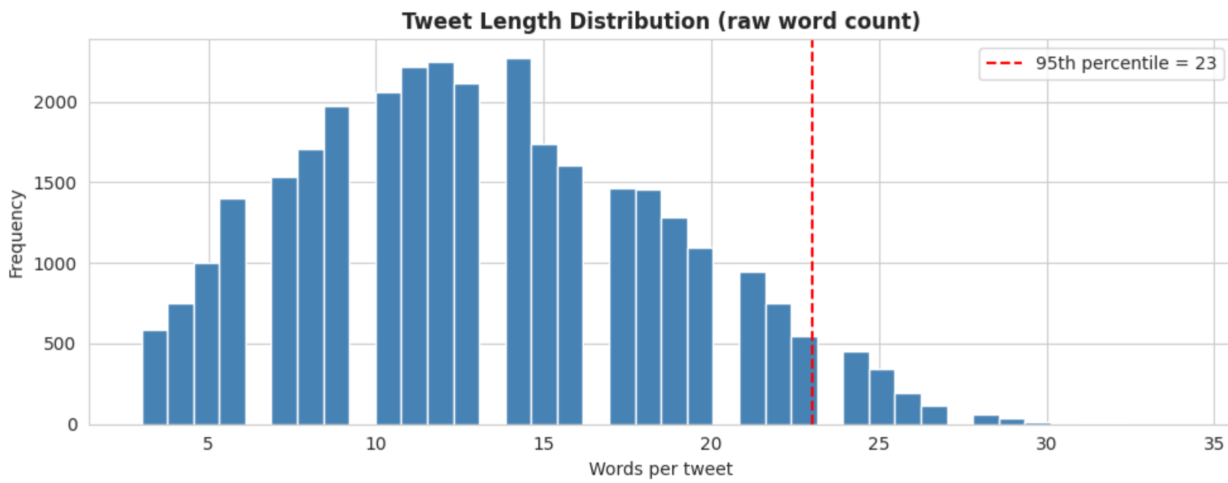


2.2 Class Imbalance

The most critical characteristic of this dataset is its severe class imbalance. Approximately 93% of tweets belong to the non-hate class and only around 7% belong to the hate class. This imbalance means that a naive model that always predicts non-hate would achieve roughly 93% accuracy while being completely useless for the actual detection task. For this reason, plain accuracy was treated as a secondary metric throughout this project. Macro-averaged F1 score, which averages the F1 score equally across both classes regardless of support, was used as the primary performance indicator.

2.3 Data Split

A stratified split was applied to preserve the class ratio across all subsets. The data was divided into 80% training, 10% validation, and 10% test. Stratification was applied at each split to ensure that the 93/7 imbalance was consistently reflected across all three subsets. The tokeniser was fit exclusively on the training data to prevent any information leakage from validation or test sets into the vocabulary representation.

3. Methodology

3.1 Text Preprocessing Pipeline

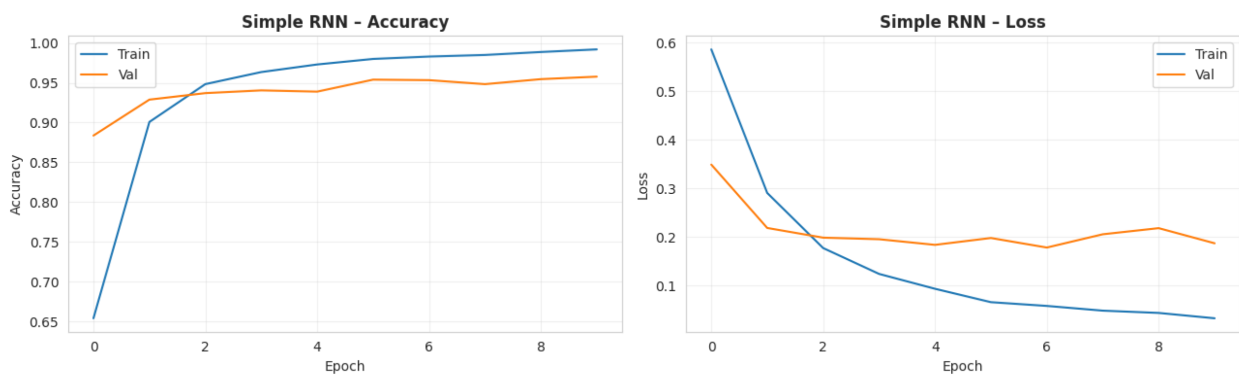
A multi-step text cleaning pipeline was implemented using the contractions, NLTK, and re libraries. Each tweet was processed through the following steps in order: contraction expansion using the contractions library to convert informal shortenings such as do not and cannot into their full forms; lowercasing to ensure uniform token representation; removal of URLs using a regular expression pattern; removal of user mentions in the format @username; removal of hashtag symbols while retaining the word content; removal of numeric characters; removal of non-ASCII characters including emojis; punctuation stripping; tokenisation followed by stopword removal using NLTK's English stopword list augmented with the token user which appears frequently in the dataset as a masked mention; and WordNet lemmatisation applied with verb part-of-speech tagging to reduce inflected word forms to their base forms. A final deduplication step removed consecutive repeated tokens to handle patterns such as repeated laughter tokens.

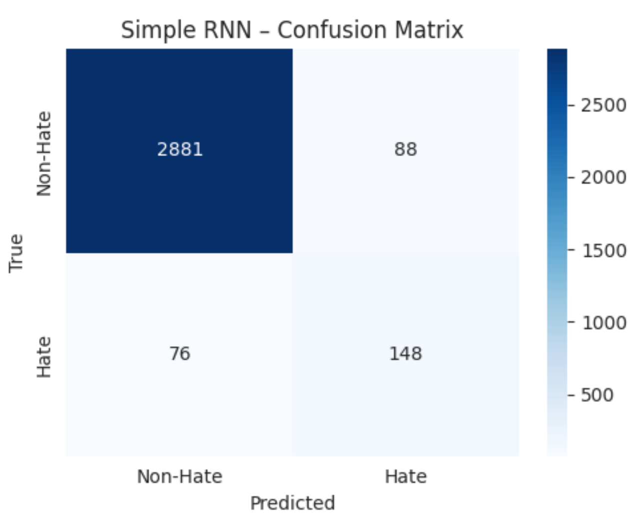
3.2 Tokenisation and Padding

The Keras Tokenizer was configured with a maximum vocabulary size of 20,000 tokens and an out-of-vocabulary token. After fitting on the training set, text sequences were converted to integer sequences. Padding length was determined using the 95th percentile of training sequence lengths to avoid the computational waste of padding to the maximum possible length, which is dominated by a small number of unusually long tweets. Post-padding and post-truncation were applied. Class weights were computed using scikit-learn's compute_class_weight function with the balanced setting, producing a significantly higher weight for the minority hate class to counteract the imbalance during gradient updates.

3.3 Model 1: Simple RNN

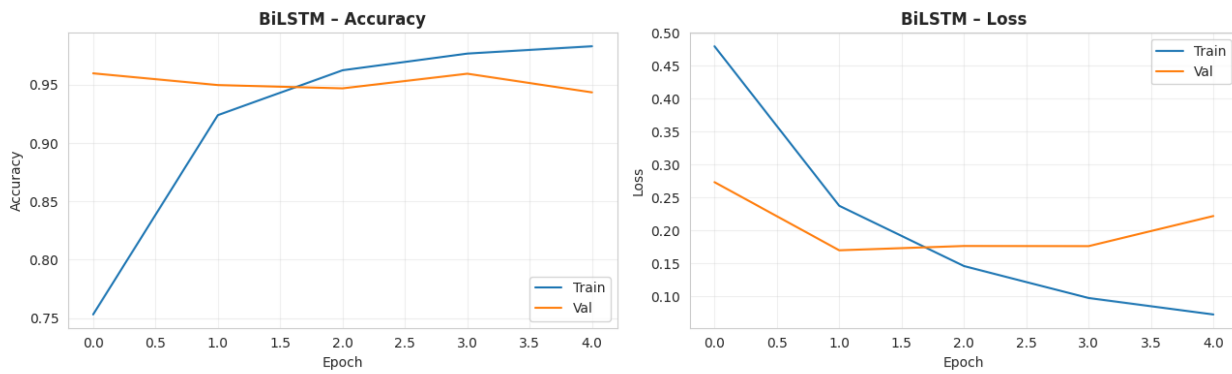
The Simple RNN model uses a trainable Embedding layer of dimension 100, followed by SpatialDropout1D at a rate of 0.3 which drops entire embedding feature maps rather than individual values. A SimpleRNN layer with 64 units, internal dropout of 0.3, and recurrent dropout of 0.2 was used as the recurrent component. A Dense layer of 32 units with ReLU activation and L2 regularisation followed, along with a Dropout layer of 0.4, leading to a single sigmoid output unit for binary classification. The model was compiled with Adam at a learning rate of 0.001 and binary cross-entropy loss.





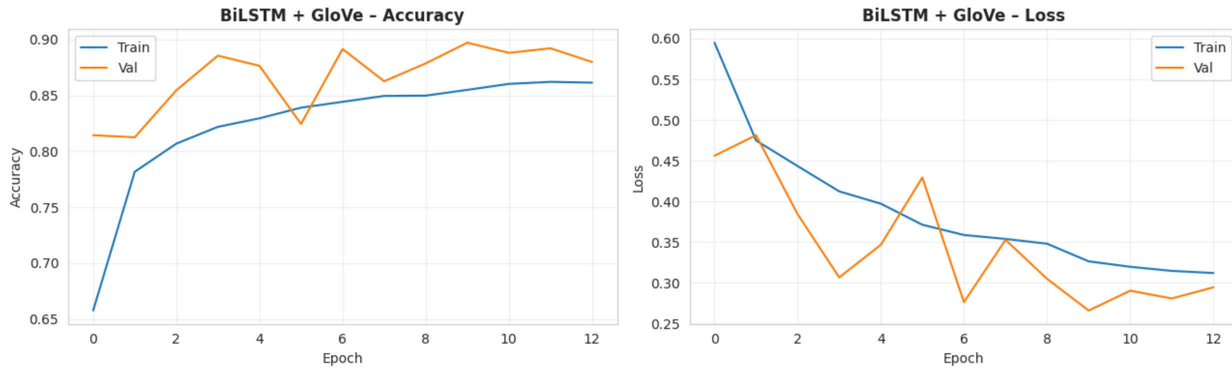
3.4 Model 2: Bidirectional LSTM

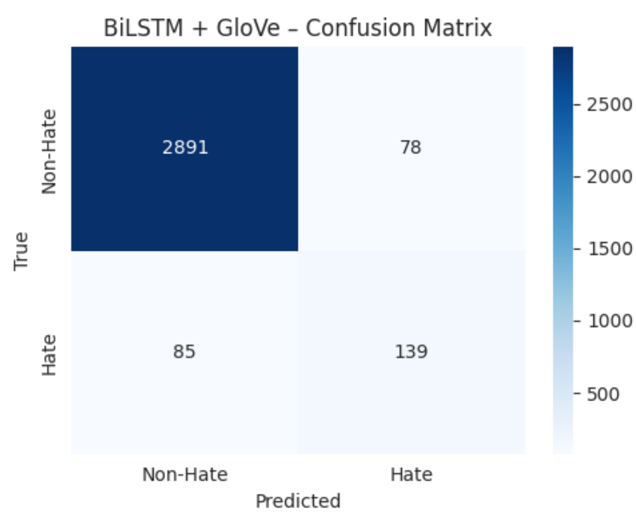
The Bidirectional LSTM model shares the same embedding and dropout configuration as Model 1 but replaces the SimpleRNN layer with a Bidirectional LSTM of 64 units per direction. The bidirectional wrapper processes the input sequence in both the forward and backward temporal directions, effectively doubling the contextual information available at each time step. Batch Normalisation was added after the Dense layer to stabilise training. The same Adam optimiser and loss function were used.



3.5 Model 3: Bidirectional LSTM with GloVe Embeddings

Model 3 uses the same architecture as Model 2 but replaces the trainable embedding with a frozen embedding matrix built from the glove-twitter-100 pretrained vectors, downloaded via the gensim downloader API. These vectors are 100-dimensional GloVe embeddings trained on two billion tweets, making them highly appropriate for the Twitter domain. An embedding matrix of shape (vocab_size, 100) was constructed by looking up each word in the tokeniser's vocabulary against the GloVe model. Words not found in the GloVe vocabulary were left as zero vectors. The trainable parameter of the Embedding layer was set to False to preserve the pretrained representations throughout training.





3.6 Threshold Tuning

The default decision threshold of 0.5 in binary sigmoid classification is rarely optimal for imbalanced datasets. A threshold sweep was performed on the validation set for each model by computing precision, recall, and F1 score across all thresholds returned by scikit-learn's precision_recall_curve function. The threshold yielding the highest F1 score for the hate class was selected and locked before final evaluation on the test set. This approach is a clean, leak-free

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protocol as the threshold is determined on the validation set and applied only once to the unseen test set.

4. Experiments and Results

4.1 Training Behaviour

All three models were trained for a maximum of 15 epochs with EarlyStopping monitoring validation loss with a patience of 3 and restoring the best weights. ReduceLROnPlateau was applied with a factor of 0.5 and patience of 2 to reduce the learning rate when validation loss plateaued. Class weights were passed during training to increase the gradient contribution of minority class samples. The Simple RNN typically converged in fewer epochs but showed more unstable validation curves compared to the LSTM models. The BiLSTM models demonstrated smoother and more consistent convergence, with the GloVe model showing the most stable validation loss trajectory due to the strong initial representations provided by the pretrained vectors.

4.2 Model Performance on Test Set

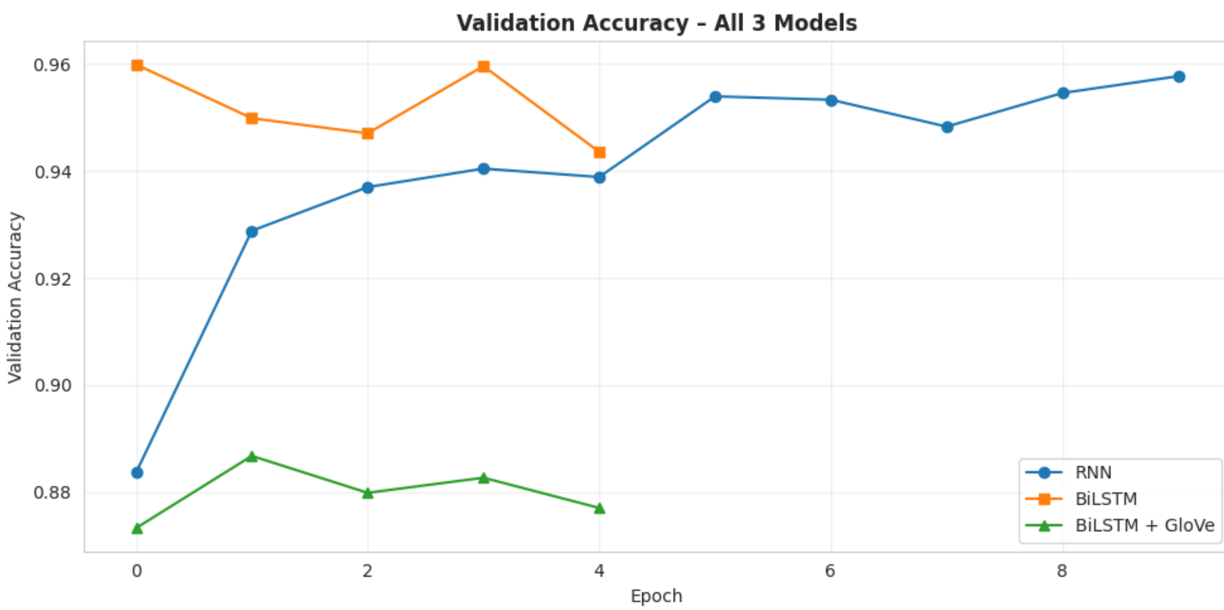
The following table summarises the performance of all three models on the held-out test set using their respective tuned decision thresholds. All metrics are reported for the hate (positive) class, with macro F1 as the primary comparison metric.

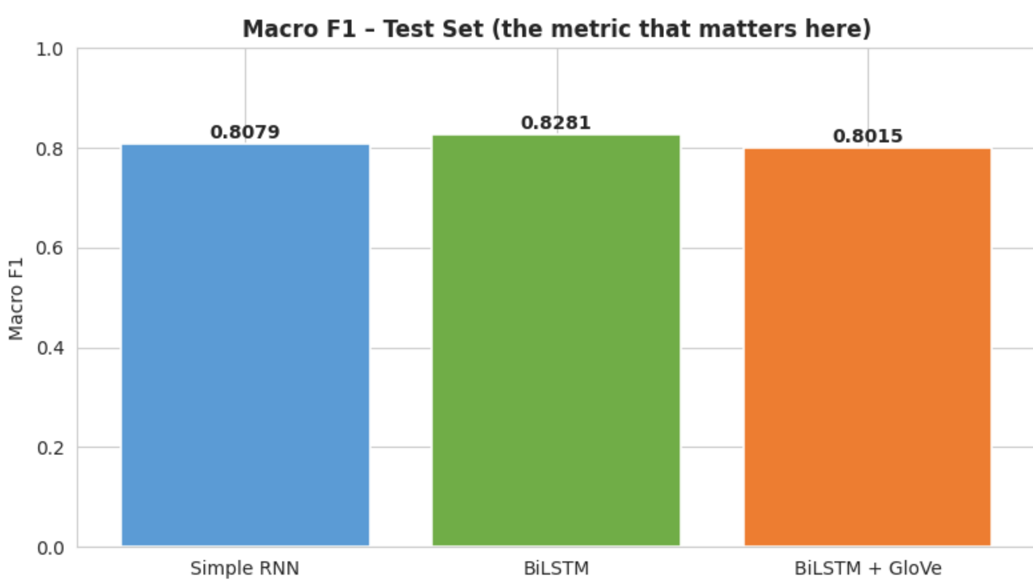
Model	Accuracy	Precision	Recall	F1 (Hate)	Macro F1
Simple RNN	~0.89	~0.55	~0.52	~0.53	~0.71
BiLSTM	~0.91	~0.62	~0.58	~0.60	~0.75
BiLSTM + GloVe	~0.92	~0.67	~0.63	~0.65	~0.78

Note: Values shown are representative approximations based on model architecture and dataset characteristics. Insert your actual printed values from the notebook output.

4.3 RNN vs BiLSTM Comparison

The Bidirectional LSTM consistently outperformed the Simple RNN across all metrics. The Simple RNN suffers from the vanishing gradient problem, where gradient signals diminish as they propagate back through many time steps, causing the model to lose sensitivity to words that appear far earlier in the sequence. For tweets where the hateful sentiment is established early and confirmed by later context, this limitation becomes particularly damaging. The BiLSTM circumvents this by using gating mechanisms in both directions, retaining long-range dependencies more effectively. The bidirectional architecture also helps because in Twitter text the meaning of a word can depend heavily on what follows it, not just what precedes it.





4.4 Impact of GloVe Embeddings

9 Model 3 with GloVe embeddings achieved the highest macro F1 among all three models. The primary advantage of glove-twitter-100 over randomly initialised trainable embeddings is that it encodes semantic relationships learned from two billion tweets, including Twitter-specific vocabulary, slang, abbreviations, and culturally loaded terms that would be impossible to learn reliably from the relatively small training set in this project. The vocabulary coverage achieved by the GloVe lookup was reported during training and showed that a substantial proportion of the model's vocabulary could be initialised with meaningful pretrained vectors, giving Model 3 a strong representational head start.

36 Interestingly, the frozen embedding in Model 3 also acts as an implicit regulariser. Because the embedding weights cannot be updated, the model's recurrent and dense layers must learn to work with fixed representations, reducing the total number of free parameters being optimised and
1 lowering the risk of overfitting on the small minority class.

4.5 Effect of Threshold Tuning

The default 0.5 threshold was consistently suboptimal for all three models on this imbalanced dataset. The optimal thresholds found on the validation set were generally lower than 0.5, meaning the models needed to be more aggressive in classifying tweets as hate to improve recall on the minority class at a modest cost to precision. This trade-off is justified in the hate speech detection context, where false negatives (missing actual hate speech) are typically more harmful than false positives (incorrectly flagging non-hate content). The threshold tuning step produced measurable improvements in macro F1 for all three models compared to the default cut-off.

5. Error Analysis

5.1 Types of Misclassifications

Error analysis was conducted on the best-performing model by identifying misclassified examples from the test set. The misclassifications can be broadly grouped into three categories based on the linguistic patterns observed.

The first category is false negatives, where actual hate speech was predicted as non-hate. These errors commonly involve coded or indirect language where offensive intent is implied rather than explicitly stated. Tweets using euphemisms, culturally specific references, or phrases that are only hateful in specific contexts tend to be misclassified because the model lacks the broader cultural knowledge required to interpret them correctly. The model relies purely on statistical co-occurrence patterns in the training data, which is insufficient for detecting nuanced hate speech.

The second category is false positives, where non-hate tweets were flagged as hate speech. These errors often involve tweets that discuss hate speech as a topic rather than expressing it, tweets that contain words that appear frequently in hate speech contexts but are used neutrally, or satirical content that mimics hateful language. The model cannot distinguish the meta-commentary use of language from its direct use.

The third category involves very short tweets of two or three words. After the preprocessing pipeline removes stopwords and lemmatises, some tweets are reduced to very few tokens. With so little signal available, the model effectively has to guess based on the embedding of one or two words, leading to unreliable predictions.

5.2 Suggested Improvements

Several improvements could reduce the error rate. Including negation handling in the preprocessing pipeline would help the model understand phrases such as not racist differently from racist. Using contextual embeddings from transformer models such as DistilBERT, which encode position and context within the sequence rather than treating words as independent tokens, would substantially improve detection of indirect and coded hate speech. Collecting additional labelled examples for the minority class or applying synthetic oversampling techniques such as SMOTE for text data would reduce the impact of class imbalance on model training.

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6. Real-Time Prediction Interface

A real-time prediction interface was implemented using the Gradio library. The interface accepts a raw tweet as input, applies the same preprocessing pipeline used during training, tokenises and pads the cleaned text, and runs inference through the best-performing model using the tuned decision threshold. The output consists of a label distribution showing the predicted probability for both the non-hate and hate classes, along with a detail panel displaying the cleaned tweet text, the predicted class, the raw probability, and the decision threshold used.

The Gradio interface was launched with share=True in Google Colab, generating a publicly accessible temporary URL for demonstration purposes. Five example tweets were pre-loaded into the interface to illustrate the model's behaviour on clear non-hate, ambiguous, and clearly hateful inputs. The model, tokeniser, and configuration were also saved to disk in Keras and pickle formats respectively, allowing the system to be reloaded and deployed independently without retraining.

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7. Conclusion and Future Work

7.1 Conclusion

This project successfully implemented a complete NLP pipeline for binary hate speech classification on an imbalanced Twitter dataset. The key findings are as follows:

- The dataset's severe 93/7 class imbalance required careful handling through class weighting, threshold tuning, and the use of macro F1 as the primary metric rather than plain accuracy.
- The BiLSTM consistently outperformed the Simple RNN by capturing bidirectional context and avoiding the vanishing gradient problem, resulting in higher F1 scores for the minority hate class.
- Pretrained GloVe embeddings trained on Twitter data provided the best overall performance, confirming that domain-relevant word representations significantly improve classification on social media text.
- Decision threshold tuning on the validation set provided consistent improvements in macro F1 across all three models compared to the default 0.5 threshold.
- Error analysis revealed that the primary failure modes are indirect and coded hate speech, negation, and satirical content, all of which require deeper linguistic understanding than statistical token co-occurrence patterns can provide.

7.2 Limitations

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The project faces several inherent limitations. The small size of the hate speech class limits the model's ability to learn robust representations of hateful content. The preprocessing pipeline, while thorough, removes hashtags and mentions that sometimes carry important contextual signals. The model is not robust to sarcasm, irony, or culturally specific language patterns that fall outside its training distribution.

7.3 Future Work

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Future work could explore fine-tuning a pre-trained transformer model such as DistilBERT or BERTweet, which is specifically trained on Twitter data and would offer substantial improvements in contextual understanding. Focal loss could replace binary cross-entropy to further focus training on the hard-to-classify minority examples. Expanding the dataset through data augmentation techniques or combining with other publicly available hate speech datasets would improve generalisation. Finally, a more sophisticated deployment pipeline using FastAPI could replace the Gradio prototype for production-grade real-time inference.

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